Centro de Investigación y de Estudios Avanzados del Instituto
Politécnico Nacional

## DEPARTAMENTO DE FÍSICA

TRABAJO FIN DE GRADO

# Sensibilidades a radio de carga de neutrinos en experimentos de neutrinos de reactor involucrando CE$\nu$NS

Realizado por
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**PHYSICS DEPARTMENT**

# Sensitivities to neutrino charge radius at reactor neutrino experiments involving CE$\nu$NS

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**In order to obtain the**

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**Under the supervision of**

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# Agradecimientos

*agradecimiento aqui.*

# Resumen

resumen aqui

# Abstract

resumen en ingles

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# 1. Introduction

intro

# 2. Electroweak theory in the Standard Model of Particles

## 2.1. Introduction

The Standard Model of particles (SM) is the most currently utilized theory of interactions of subatomic particles. It describes all matter and forces between them (excluding gravity) as interactions between 12 fermions (6 quarks and 6 leptons) with their corresponding anti-fermions, 4 spin-1 gauge bosons and 1 scalar boson as shown in Fig. 2.1.

## 2.2. Symmetries and fermion content

## 2.3. Massive and Massless Mediators

### 2.3.1 Electroweak mixing $\sin^2 theta_W$

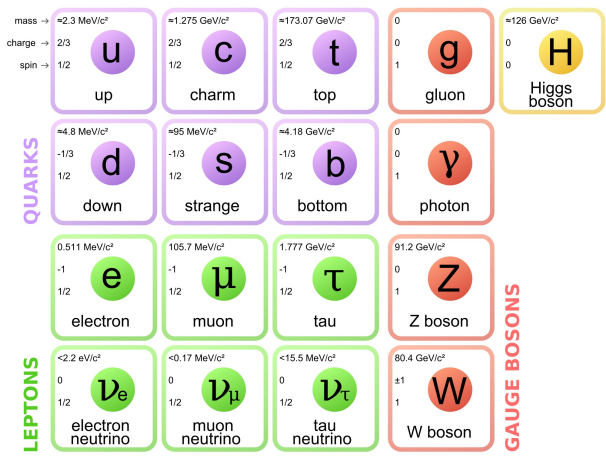


***Figure 2.1:*** *Particles in the SM*

# 3. Radiative Corrections and Neutrino Charge Radius

## 3.1. Introduction

## 3.2. Neutrino Charge Radius

## 3.3. One-loop Corrections to the $\nu\nu\gamma$ vertex

## 3.4. NCR as a radiative correction of Electroweak Mixing

# 4. Coherent Elastic Neutrino-Nucleus Scattering

## 4.1. Introduction

## 4.2. Cross Section

## 4.3. Experiments

## 4.4. Quenching Factors

## 4.5. Neutrino Sources

## 4.6. Form Factors

# 5. SBC-ININ-LAr

## 5.1. Results

# 6. Summary and conclusions